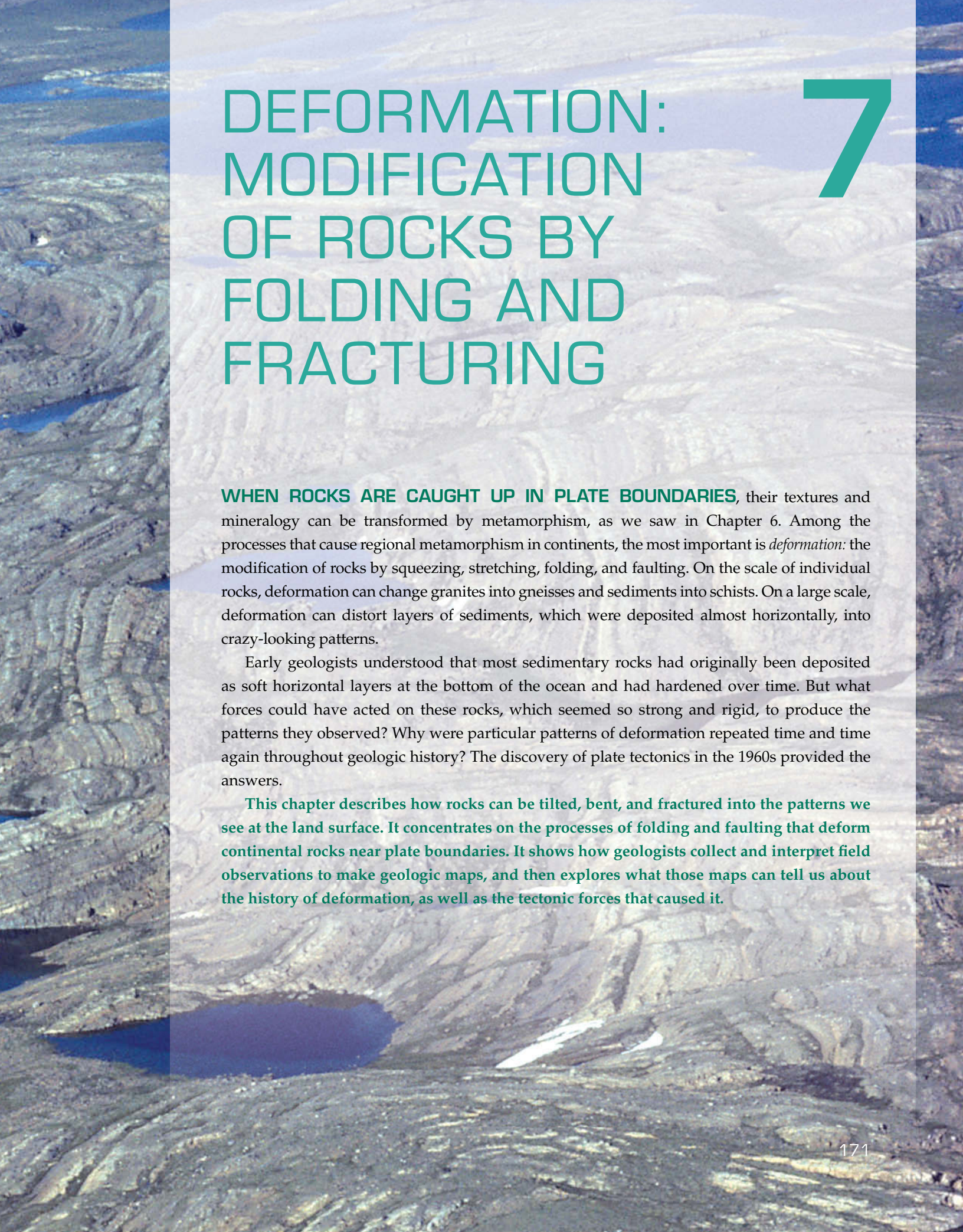




DEFORMATION: MODIFICATION OF ROCKS BY FOLDING AND FRACTURING

7

WHEN ROCKS ARE CAUGHT UP IN PLATE BOUNDARIES, their textures and mineralogy can be transformed by metamorphism, as we saw in Chapter 6. Among the processes that cause regional metamorphism in continents, the most important is *deformation*: the modification of rocks by squeezing, stretching, folding, and faulting. On the scale of individual rocks, deformation can change granites into gneisses and sediments into schists. On a large scale, deformation can distort layers of sediments, which were deposited almost horizontally, into crazy-looking patterns.

Early geologists understood that most sedimentary rocks had originally been deposited as soft horizontal layers at the bottom of the ocean and had hardened over time. But what forces could have acted on these rocks, which seemed so strong and rigid, to produce the patterns they observed? Why were particular patterns of deformation repeated time and time again throughout geologic history? The discovery of plate tectonics in the 1960s provided the answers.

This chapter describes how rocks can be tilted, bent, and fractured into the patterns we see at the land surface. It concentrates on the processes of folding and faulting that deform continental rocks near plate boundaries. It shows how geologists collect and interpret field observations to make geologic maps, and then explores what those maps can tell us about the history of deformation, as well as the tectonic forces that caused it.

■ Plate Tectonic Forces

Deformation is a general term that encompasses the folding, faulting, shearing, compression, and extension of rock by plate tectonic forces. The types of deformation we see exposed at Earth's surface are caused mainly by the horizontal movements of the lithospheric plates relative to one another. For this reason, the tectonic forces that deform rocks at plate boundaries are predominantly *horizontally directed* and depend on the direction of relative plate movement:

- **Tensional forces**, which stretch and pull rock formations apart, dominate at divergent boundaries, where plates move away from each other.
- **Compressive forces**, which squeeze and shorten rock formations, dominate at convergent boundaries, where plates move toward each other.
- **Shearing forces**, which shear two parts of a rock formation in opposite directions, dominate at transform-fault boundaries, where plates slide past each other.

If plates were perfectly rigid, the plate boundaries would be sharp lineations, and points on either side of those boundaries would move at the relative plate velocity. This idealization is often a good approximation in the oceans, where rift valleys at mid-ocean ridges, deep-sea trenches, and nearly vertical transform faults form narrow plate boundary zones, often just a few kilometers wide.

Within continents, however, the deformation caused by plate movements can be “smeared out” across a plate boundary zone hundreds or even thousands of kilometers wide. The continental crust does not behave rigidly within these broad zones, so rocks at the surface are deformed by folding and faulting. Folds in rocks are like folds in clothing. Just as cloth pushed together from opposite sides bunches up in folds, layers of rock slowly compressed by tectonic forces in the crust can be pushed into folds (Figure 7.1a). Tectonic forces can also cause a rock formation to break and slip on both sides of a fracture, forming a fault (Figure 7.1b). When such a break occurs suddenly, the result is an earthquake. Active zones of continental deformation are marked by frequent earthquakes.

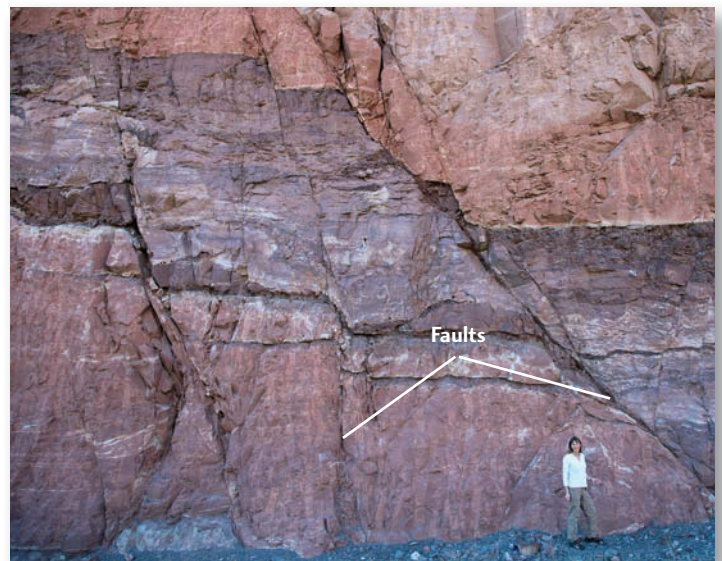
Geologic folds and faults can range in size from centimeters to meters (as in Figure 7.1) to tens of kilometers or more. Many mountain ranges are actually a series of large folds and faults that have been weathered and eroded. From the geologic record of deformation laid out on Earth's surface, geologists can deduce the directions of movement at ancient plate boundaries and reconstruct the tectonic history of the continental crust.

■ Mapping Geologic Structure

Faults and folds are examples of the basic features geologists observe and map to reconstruct crustal deformation. To better understand this process, we need information



(a)



(b)

FIGURE 7.1 ■ Rocks subjected to tectonic forces are deformed by folding and faulting. (a) An outcrop of originally horizontal rock layers has been bent into folds by compressive tectonic forces. (b) An outcrop of once-continuous rock layers has been displaced along small faults by tensional tectonic forces. [(a) Tony Waltham; (b) Marli Bryant Miller.]



FIGURE 7.2 ■ Dipping limestone and shale beds on the coast of Somerset County, England. The children are walking along the strike of the beds. The beds dip to the left at an angle of about 15° . [Chris Pellant.]

about the geometry of faults and folds. The best place to find this information is at an *outcrop*, where the solid rock that underlies the ground surface—the *bedrock*—is exposed (not obscured by vegetation, soil, or loose boulders). At an outcrop, geologists can identify distinct **formations**: groups of rock layers that can be identified throughout a region by their physical properties. Some formations consist of a single rock type, such as limestone. Others are made up of thin, interlayered beds of different kinds of rock, such as sandstone and shale. However they vary, each formation comprises a distinct set of rock layers that can be recognized and mapped as a unit.

Figure 7.1a shows an outcrop in which the folding of sedimentary beds is clearly visible. Often, however, folded rocks are only partly exposed in an outcrop and can be seen only as inclined layers (**Figure 7.2**). The orientation of those layers is an important clue we can use to piece together a picture of the overall geologic structure. Two measurements describe the orientation of a rock layer exposed at an outcrop: the strike and the dip of the layer's surface.

Measuring Strike and Dip

The **strike** is the compass direction of a rock layer where it intersects with a horizontal surface. The **dip**, which is measured at right angles to the strike, is simply the amount of tilting—the angle at which the rock layer inclines from the horizontal. **Figure 7.3** shows how strike and dip are measured in the field. A geologist might describe the outcrop in this figure as “a bed of coarse-grained sandstone striking east-west and dipping 45° south.”

Geologic Maps

Geologic maps are two-dimensional representations of the rock formations exposed at Earth's surface (**Figure 7.4**).

To create a geologic map, a geologist must choose an appropriate *scale*—the ratio of distance on the map to true surface distance. A common scale for geologic field mapping is 1:24,000 (pronounced “one to twenty-four thousand”), which means that 1 inch on the map corresponds to 24,000 inches (2000 feet) on Earth's surface. To depict the geology of an entire state, a geologist would choose a smaller scale: say, 1:1,000,000, where 1 cm represents 10 km and 1 inch almost 16 miles. The smaller the scale, the less detail can be depicted on the map.

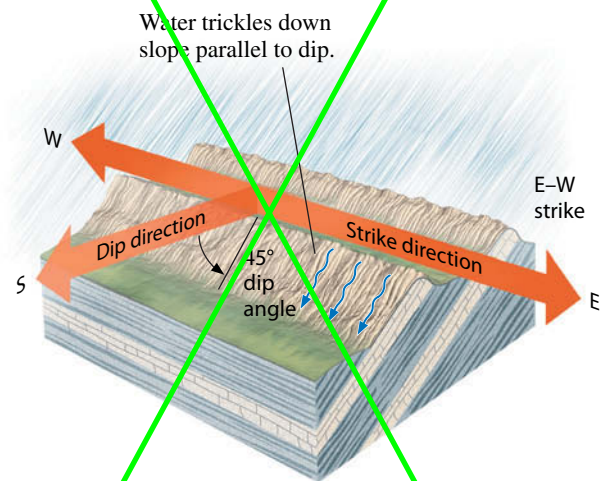


FIGURE 7.3 ■ The strike and dip of a rock layer define its orientation at a particular place. The strike is the compass direction of a rock layer along the line of its intersection with a horizontal surface. The dip is the angle and direction of the steepest descent of a rock layer from the horizontal, measured at right angles to the strike. Here, the strike is east-west and the dip is 45° to the south. [After A. Maltman, *Geological Maps: An Introduction*, 2nd ed. New York: Van Nostrand Reinhold, 1998.]

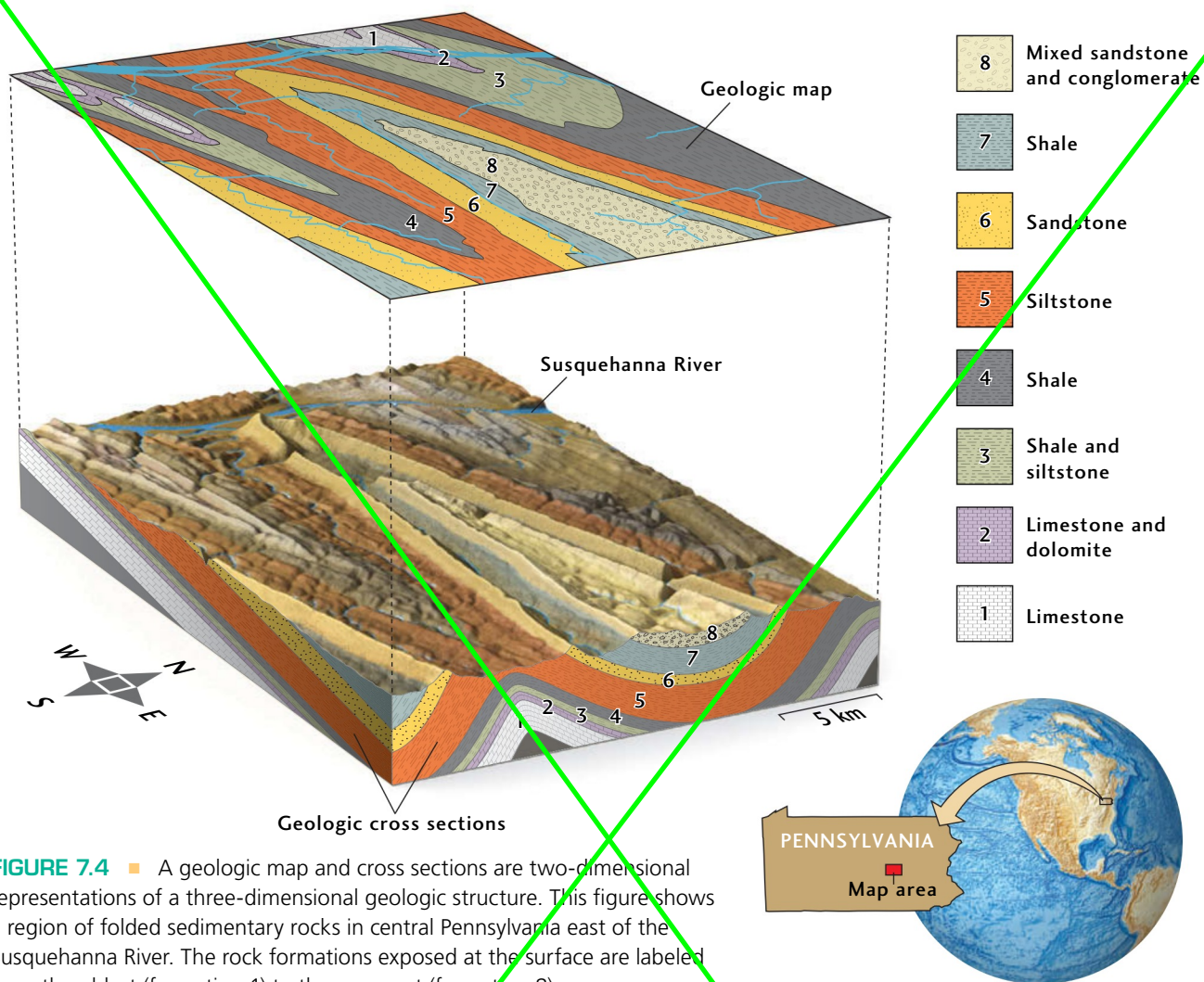


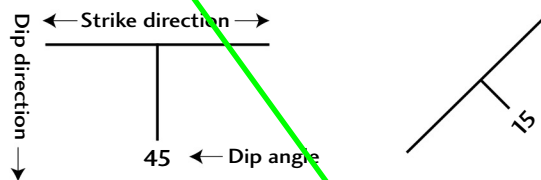
FIGURE 7.4 ■ A geologic map and cross sections are two-dimensional representations of a three-dimensional geologic structure. This figure shows a region of folded sedimentary rocks in central Pennsylvania east of the Susquehanna River. The rock formations exposed at the surface are labeled from the oldest (formation 1) to the youngest (formation 8).

Geologists keep track of different rock formations by assigning each formation a particular color on the map, usually keyed to the rock's type and age (see Figure 7.4). Many different rock formations may be exposed in highly deformed regions, so geologic maps can be very colorful!

Softer rocks, such as mudstones and other poorly consolidated sediments, are more easily eroded than harder rocks, such as limestones, sandstones, or metamorphic rocks. Consequently, rock types can exert a strong influence on the topography of the land surface and the exposure of rock formations (see Figure 7.4). The important relationships between geology and topography can be made clear by plotting the contours of the land surface on a geologic map.

Because geologic maps can represent such a huge amount of information, they have been called “textbooks on a piece of paper.” To convey this information more concisely, geologic maps are annotated with special symbols that indicate the local strike and dip of rock formations and with special types of lines that mark faults and other significant features. For instance, the strike and dip of

rock formations are indicated on a geologic map by T-like symbols:



The top of the T indicates the strike direction, the shank of the T indicates the dip direction, and the number gives the dip angle in degrees. On a map where north is up, the symbol on the left thus describes the sandstone bed in Figure 7.3, which has an east-west strike and a dip of 45° to the south. The one on the right describes a formation that strikes northeast-southwest and dips to the southeast at an angle of 15° , such as the beds in Figure 7.2.

Of course, not every detail of surface geology can be represented on a map, so geologists must simplify the

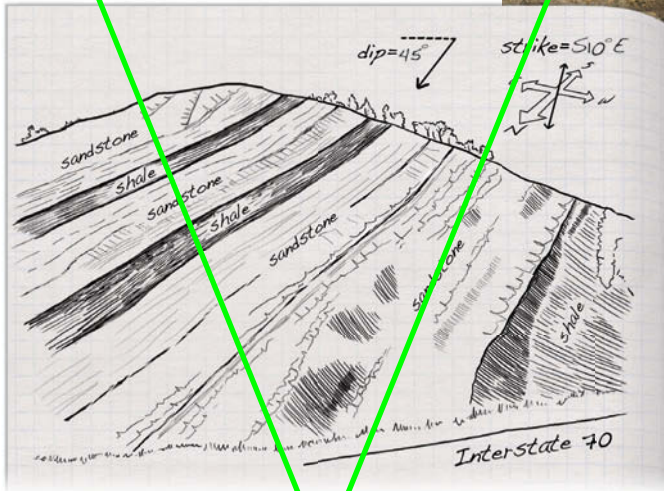


FIGURE 7.5 ■ Geologic cross sections can sometimes be observed directly in the field. This road cut, which is part of the Lariat Loop Scenic Byway in Colorado, shows a near-vertical cross section through a sequence of sedimentary beds tilted by the uplift of the Rocky Mountains. [James Steinberg/Science Source.]

structures they see, perhaps by representing a complex zone of faulting as a single fault or by ignoring folds too small to show at the scale they have chosen. They may also “dust off” their maps by ignoring thin layers of soil and loose rock that cover up the geologic structure, portraying the structure as if outcrops existed everywhere. You should therefore think of a geologic map as a simplified *scientific model* of the surface geology.

Geologic Cross Sections

Once a region has been mapped, the two-dimensional geologic map must be interpreted in terms of the underlying three-dimensional geologic structure. How can the shapes of the rock layers be reconstructed, even when erosion has removed parts of a formation? The process is like putting together a three-dimensional jigsaw puzzle with some of the pieces missing. Common sense and intuition play important roles, as do basic geologic principles.

To piece together the puzzle, geologists construct **geologic cross sections**—diagrams showing the features that would be visible if vertical slices were made through part of the crust. Some small cross sections can actually be seen in the vertical faces of cliffs, quarries, and road cuts (**Figure 7.5**). Cross sections spanning much larger areas can be constructed from the information on a geologic map, including the strikes and dips observed at outcrops. The accuracy of cross sections based on surface mapping

can be improved by drilling boreholes to collect rock samples as well as by seismic imaging. But drilling and seismic imaging are expensive, so data collected by these methods are usually available only for areas that have been explored for oil, water, or other valuable natural resources.

Figure 7.4 shows a geologic map of an area where originally horizontal beds of sedimentary rock were bent into a series of folds and eroded into a set of zigzagging ridges and valleys. We will explore some of the geologic relationships seen in this map later in this chapter. But first we will investigate the basic processes by which rocks deform.

■ How Rocks Deform

Rocks deform in response to the tectonic forces acting on them. Whether they will respond to tectonic forces by folding, faulting, or some combination of the two depends on the orientation of the forces, the rock type, and the physical conditions (such as temperature and pressure) during deformation.

Brittle and Ductile Behavior of Rocks in the Laboratory

In the mid-1900s, geologists began to explore the forces of deformation by using powerful hydraulic rams to bend

This sample was compressed under conditions representative of the shallow crust. The fractures indicate that the marble is brittle at the laboratory equivalent of shallow depth.

This sample was compressed under conditions representative of the deeper crust. It deformed by bulging smoothly, indicating that marble is ductile at greater depth.

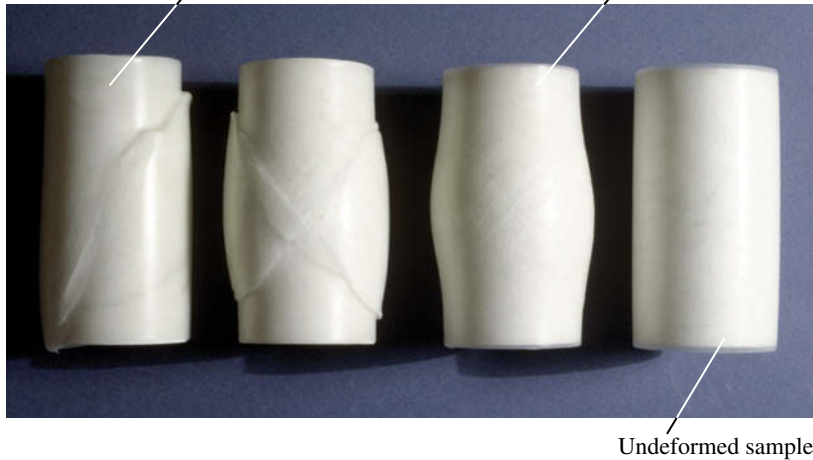


FIGURE 7.6 ■ Results of laboratory experiments conducted to discover how rock—in this case, marble—is deformed by compressive forces. The marble samples are encased in translucent plastic jackets, which explains their shiny appearance. [Fred and Judith Chester/John Handin Rock Deformation Laboratory of the Center for Tectonophysics.]

and break small samples of rock. Engineers had invented such machines to measure the strength of concrete and other building materials, but geologists modified them to discover how rocks deform at pressures and temperatures high enough to simulate physical conditions deep in Earth's crust.

In one such experiment, the researchers applied compressive force by pushing down with a hydraulic ram on one end of a small cylinder of marble while at the same time maintaining confining pressure on the cylinder (Figure 7.6). Under low confining pressures, equivalent to those found at shallow depths in Earth's crust, the marble sample deformed only a small amount until the compressive force on its end was increased to the point that the entire sample suddenly broke (see Figure 7.6, left side). This experiment showed that marble behaves as a **brittle** material at the low confining pressures found in the shallow crust. Repeating the experiment under high confining pressures, equivalent to those that often accompany metamorphism, produced a different result: the marble sample slowly and steadily deformed into a shortened, bulging shape without fracturing (see Figure 7.6, right side). Marble thus behaves as a pliable, or **ductile**, material at the high confining pressures found deep in the crust.

Other experiments showed that when marble was heated to temperatures as high as those that accompany metamorphism, it acted as a ductile material at a lower confining pressure—just as heating wax changes it from a hard material that can break into a soft material that flows. The researchers concluded that the particular marble they were working with would deform by faulting at depths shallower than a few kilometers, but would deform by folding at the greater crustal depths where metamorphism generally occurs.

Brittle and Ductile Behavior of Rocks in Earth's Crust

Natural conditions in Earth's crust cannot be reproduced exactly in the laboratory. Tectonic forces are applied over millions of years, whereas laboratory experiments are rarely conducted over more than a few hours, or at most a few weeks. Nevertheless, the results of laboratory experiments can help us interpret what we see in the field. Geologists keep the following points in mind when mapping crustal folds and faults:

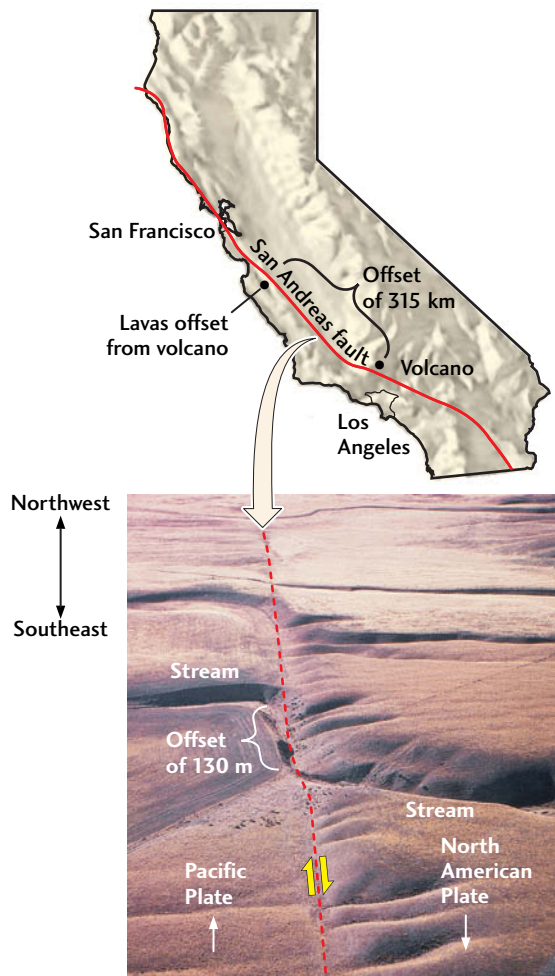
- The same rock can be brittle at shallow depths (where temperatures and pressures are relatively low) and ductile deep in the crust (where temperatures and pressures are higher). Metamorphism is usually accompanied by ductile deformation.
- Rock type affects deformation. In particular, the hard igneous and metamorphic rocks that form the crystalline *basement* of a continent (the crust beneath layers of sediments) often behave as brittle materials, fracturing along faults during deformation, while softer sedimentary rocks that overlie them often behave as ductile materials, folding gradually.
- A rock formation that would behave as a ductile material if deformed slowly may behave as a brittle material if deformed more rapidly. (Think of Silly Putty, which flows as a ductile clay when you squeeze it slowly, but breaks into pieces when you pull it apart very quickly.)
- Rocks break more easily when subjected to tensional (pulling and stretching) forces than when subjected to compressive forces. Sedimentary rock formations that will deform by folding during compression will often break along faults when subjected to tension.

■ Basic Deformation Structures

Geologists use the simple geometric concepts and measurements described earlier in this chapter to classify features such as faults and folds into different types of deformation structures.

Faults

A **fault** is a fracture that displaces the rock on either side of it. We can measure the orientation of the fracture surface, or *fault surface*, by its strike and dip, just as we do for other geologic surfaces (see Figure 7.3). The movement of the block of rock on one side of the fault with respect to that on the other side can be described by a *slip direction* and by the total displacement, or *offset*. For small faults, such as the ones pictured in Figure 7.1b, the offset might be only a couple of meters, whereas the offset along a major transform fault, such as the San Andreas fault, can amount to hundreds of kilometers (Figure 7.7).



Rocks on either side of a fault cannot interpenetrate one another, and at high pressures below the surface, they cannot open up, so the slip direction during faulting must be parallel to the fault surface. Faults can therefore be classified by their slip direction along this surface (Figure 7.8). A **dip-slip fault** is one on which there has been relative movement of blocks of rock up or down the dip of the fault plane. A **strike-slip fault** is a fault on which the movement of blocks has been horizontal, parallel to the strike of the fault plane. When blocks of rock move along the strike and simultaneously up or down the dip, the result is an *oblique-slip fault*. Dip-slip faults are caused by compressive or tensional forces, whereas strike-slip faults are the work of shearing forces. An oblique-slip fault results from shearing in combination with either compression or tension.

These fault types require further classification because the movement of rock can be up or down, or right or left. To describe these movements, geologists borrow some terminology used by miners, calling the block of rock above a dipping fault plane the **hanging wall** and the block of rock below it the **foot wall**. A dip-slip fault is called a **normal fault** if the hanging wall moves downward relative to the foot wall, extending the structure horizontally (Figure 7.8a). A dip-slip fault is called a *reverse fault* if the hanging wall moves upward relative to the foot wall, causing a shortening of the structure (Figure 7.8b)—the reverse of what geologists have (somewhat arbitrarily) chosen as “normal.” A **thrust fault** is a low-angled reverse fault—that is, one with a dip of less than 45° , so that the movement is more horizontal than vertical (Figure 7.8c). When subjected to horizontal compression, brittle rocks of the continental crust usually break along thrust faults with dips of 30° or less, rather than along more steeply dipping reverse faults.

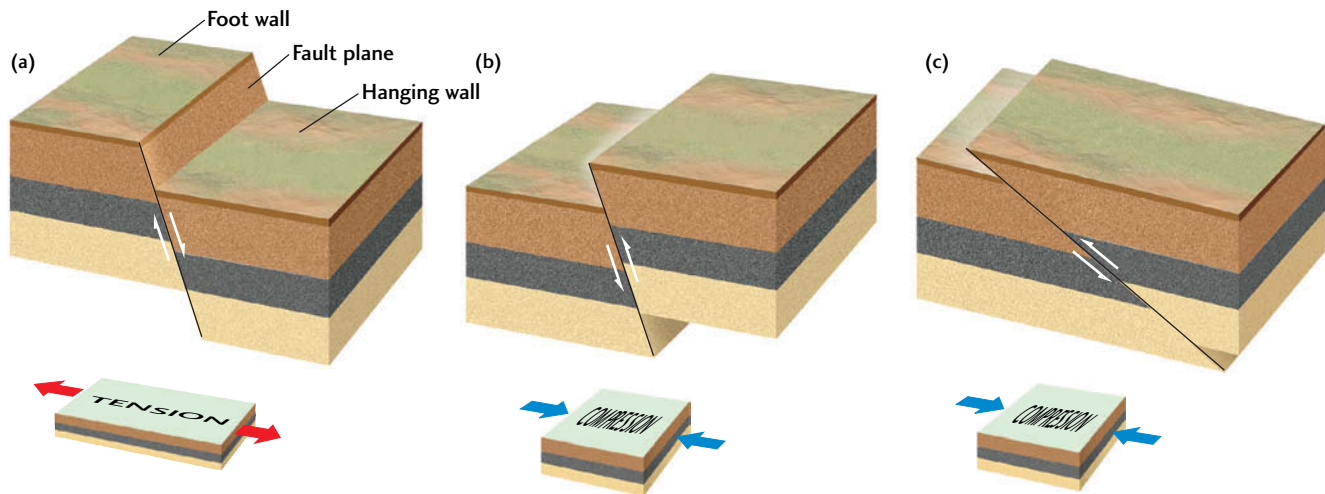
A strike-slip fault is a *left-lateral fault* if an observer on one side of the fault sees that the block on the opposite side has moved to the left (Figure 7.8d). It is a *right-lateral fault* if the block on the opposite side appears to have moved to the right (Figure 7.8e). As you can tell from the stream offset in Figure 7.7, the San Andreas fault is a right-lateral transform fault. Other faults show both strike-slip and dip-slip motions. These faults are known as **oblique slip faults** (Figure 7.8f).

Geologists can recognize faults in the field in several ways. A fault may form a *scarp* (a cliff) that marks where

FIGURE 7.7 ■ View of the San Andreas fault, showing the northwestward movement of the Pacific Plate with respect to the North American Plate. The map shows a formation of volcanic rocks 23 million years old that has been displaced by 315 km. The fault runs from top to bottom (dashed line) near the middle of the photograph. Note the offset of the stream (Wallace Creek) by 130 m as it crosses the fault.

[University of Washington Libraries, Special Collections, John Shelton Collection, KCN7-23.]

DIP-SLIP FAULTING

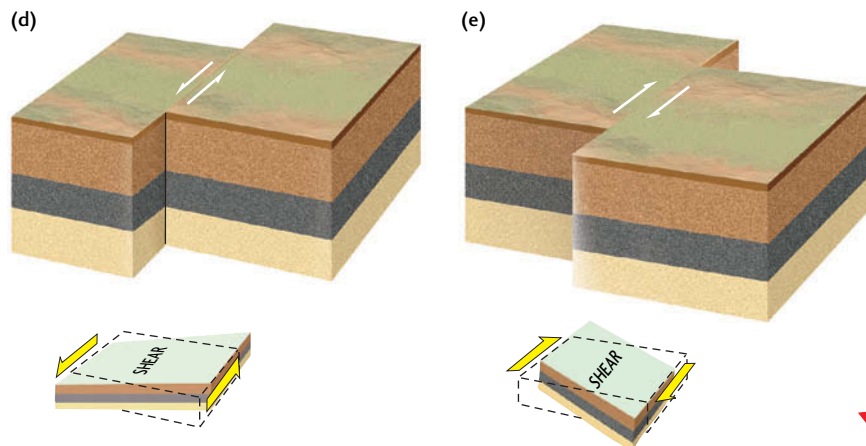


Normal faulting is caused by tensional forces that stretch a rock and tend to pull it apart.

Reverse faulting is caused by compressive forces that squeeze and shorten a rock.

A thrust fault is a reverse fault with a shallow-dipping fault plane.

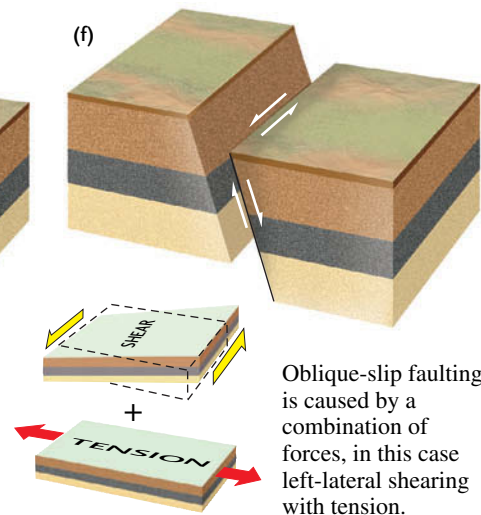
STRIKE-SLIP FAULTING



Left-lateral strike-slip fault

Right-lateral strike-slip fault

OBLIQUE-SLIP FAULTING



Oblique-slip faulting is caused by a combination of forces, in this case left-lateral shearing with tension.

FIGURE 7.8 The orientation of tectonic forces determines the style of faulting. Dip-slip faulting (a–c) is caused by compressive or tensional forces. Strike-slip faulting (d, e) is caused by shearing forces. Oblique-slip faulting (f) is caused by a combination of shearing and compressive or tensional forces.

the fault intersects the ground surface (Figure 7.9). If the offset has been large, as it is for transform faults such as the San Andreas, the rock formations facing each other across the fault may differ in type and age. When movements are smaller, offset features can be observed and measured. (As an exercise, try to match up the beds offset by the small faults in Figure 7.1b.) In establishing the time of faulting, geologists apply a simple rule: a fault must be younger than the youngest rocks it cuts (the rocks had to be there before they could break!) and older than the oldest undeformed formation that covers it.

On geologic maps, faults are represented by *fault traces*: lines that indicate the point where a fault intersects the ground surface. Normal faults are distinguished from thrust faults by the different types of “teeth” that annotate the fault trace:

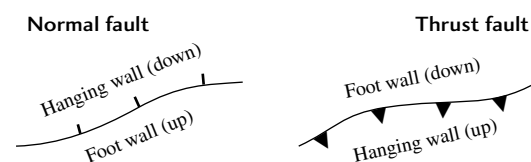




FIGURE 7.9 ■ This fault scarp is a fresh surface feature that formed by normal faulting during the 1954 Fairview Peak earthquake in Nevada.

[Garry Hayes/Geotripper Images.]

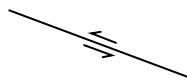


For both types of dip-slip faults, the teeth point toward the hanging wall. Examples of normal faults represented this way are shown in Figure 7.20; examples of thrust faults are shown in Figure 7.22. For strike-slip faults, the direction of movement, right-lateral or left-lateral, is indicated by a pair of arrows (yellow) that bracket the fault trace (see Figure 7.7):

Right-lateral
strike-slip fault



Left-lateral
strike-slip fault



Folds

Folding is a common form of deformation observed in layered rocks (as in Figure 7.1a). **Folds** occur when an originally planar structure, such as a sedimentary bed, is bent into a curved structure. The bending can be produced by either horizontally or vertically directed forces in the crust, just as either pushing together the opposite edges of a piece of paper or pushing up or down on one side or the other can fold it.

Like faults, folds come in all sizes. In many mountain belts, majestic, sweeping folds can be traced over many kilometers (Figure 7.10). On a much smaller scale, very thin

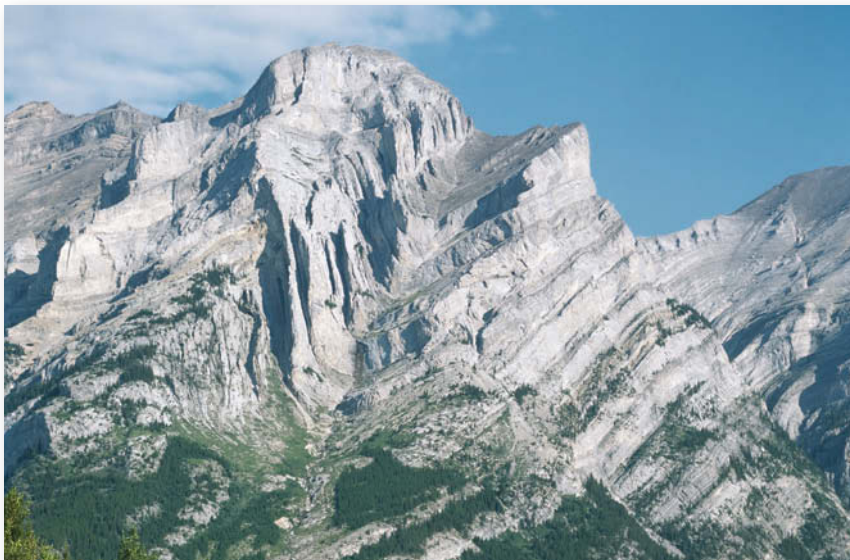


FIGURE 7.10 ■ Large-scale folds in the sedimentary rocks that form the Kananaskis mountain range in Alberta, Canada. [Photoshot.]



FIGURE 7.11 ■ Small-scale folds in sedimentary beds of anhydrite (light) and shale (dark) in West Texas. [John Grotzinger/Ramón Rivera-Moret/Harvard Mineralogical Museum.]

sedimentary beds can be crumpled into folds a few centimeters long (Figure 7.11). The bending can be gentle or severe, depending on the magnitude of the applied forces, the length of time over which they were applied, and the resistance of the rocks to deformation.

Folds in which layered rocks are bent upward into arches are called **anticlines**; those in which rocks are bent downward into troughs are called **synclines** (Figure 7.12).

The two sides of a fold are its *limbs*. The *axial plane* of a fold is an imaginary surface that divides the fold as symmetrically as possible, with one limb on either side of the plane. The line made by the lengthwise intersection of the axial plane with the rock layers is the *fold axis*. A symmetrical horizontal fold has a horizontal fold axis and a vertical axial plane with limbs dipping symmetrically away from the axis.

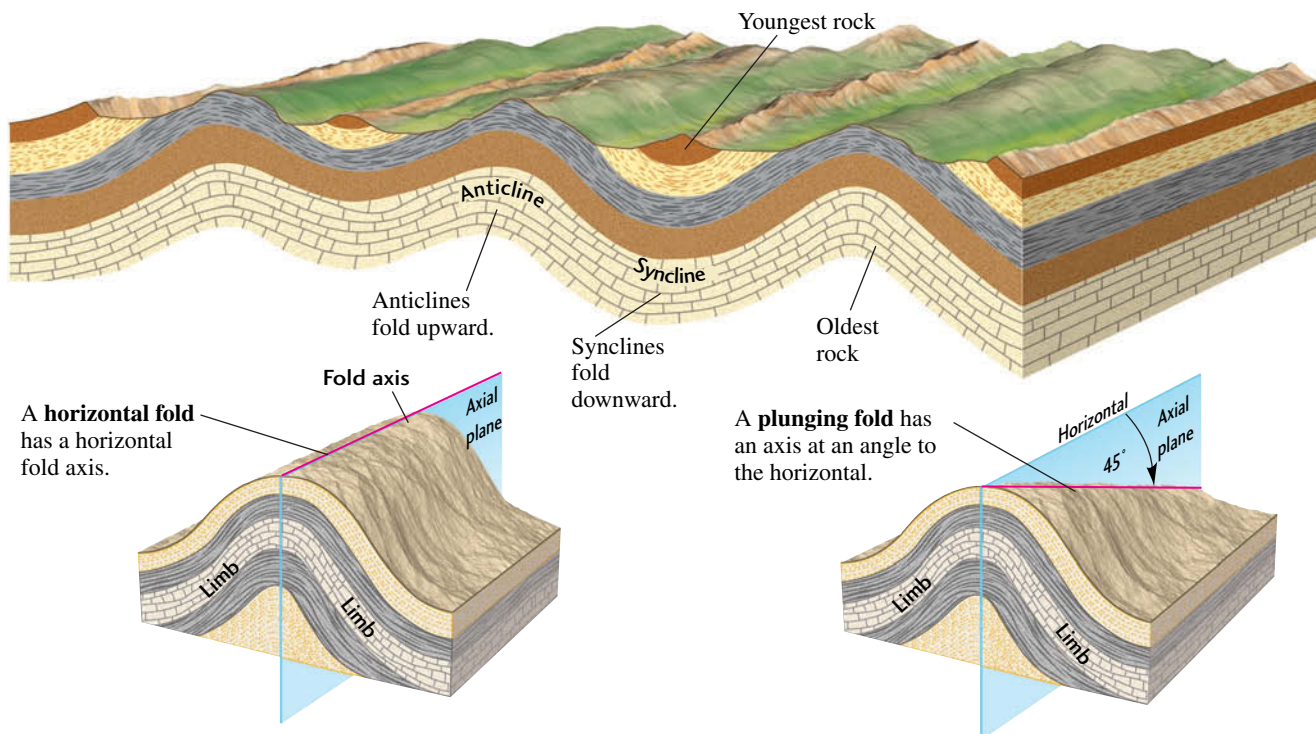


FIGURE 7.12 ■ The folding of rock layers is described by the direction of folding (upward or downward) and by the orientation of the fold axis and the axial plane.

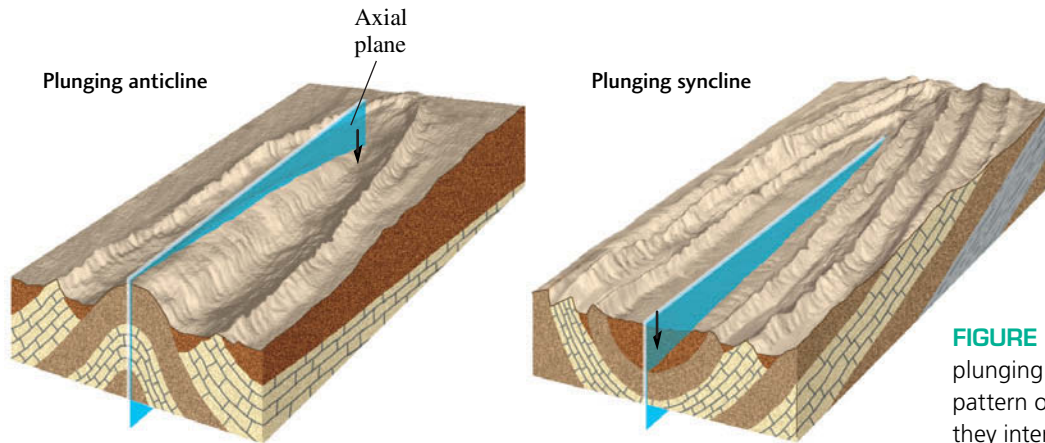


FIGURE 7.13 ■ The geometry of plunging folds. Note the converging pattern of the layers of rock where they intersect the land surface.

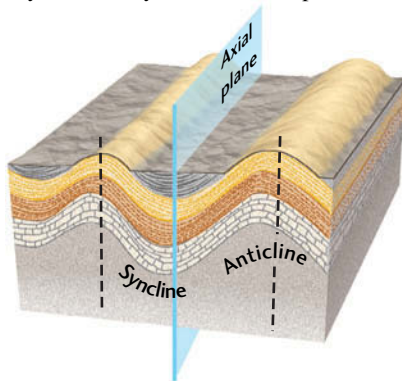
Folds rarely stay horizontal, however. Follow the axis of any fold in the field, and sooner or later the fold dies out or appears to plunge into the ground. If a fold's axis is not horizontal, it is called a *plunging fold*. **Figure 7.13** diagrams the geometry of plunging anticlines and plunging synclines. In eroded mountain belts, a zigzag pattern of outcrops may appear in the field after erosion has removed much of the surface rock from the folds. The geologic map in Figure 7.4 shows this characteristic pattern.

Nor do folds usually remain symmetrical. With increasing amounts of deformation, folds can be pushed

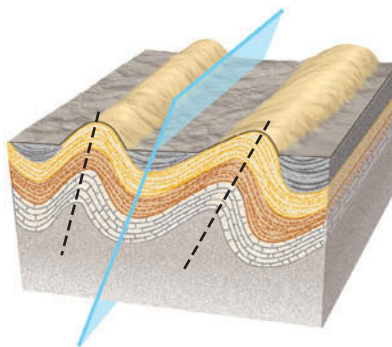
into asymmetrical shapes, with one limb dipping more steeply than the other (**Figure 7.14**). Such *asymmetrical folds* are common. When the deformation is so intense that one limb has been tilted beyond the vertical, the fold is called an *overturned fold*. Both limbs of an overturned fold dip in the same direction, but the order of the layers in the bottom limb is precisely the reverse of their original sequence—that is, older rocks are on top of younger rocks.

Observations in the field seldom provide complete information about folds. Bedrock may be obscured by

Symmetrical folds have limbs that dip symmetrically from the axial plane.



Asymmetrical folds have one limb that dips more steeply than the other.



Overturned folds have limbs that dip in the same direction, but one limb has been tilted beyond the vertical.

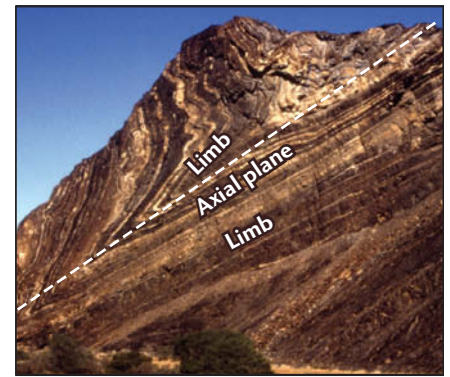
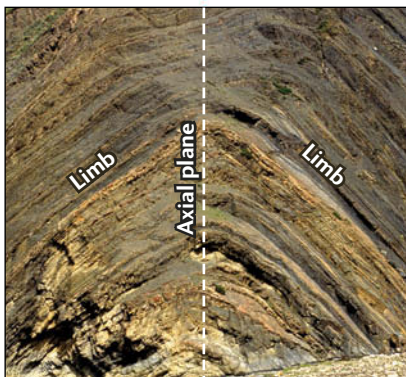
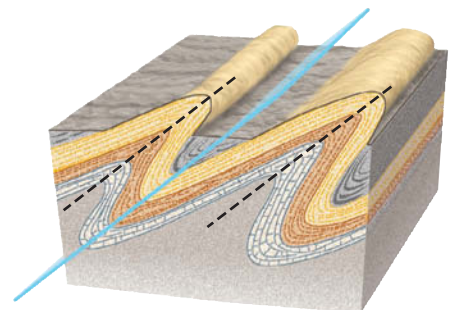


FIGURE 7.14 ■ With increasing deformation, folds are pushed into asymmetrical shapes. [(left) Tony Waltham; (center) © Photoshot; (right) Courtesy of John Grotzinger.]

overlying soils, or erosion may have removed much of the evidence of former structures. So geologists search for clues they can use to work out the relationship of one bed to another. For example, in the field or on a geologic map, an eroded anticline might be recognized as a strip of older rocks forming a core bordered on both sides by younger rocks dipping away from the core. An eroded syncline might appear as a core of younger rocks bordered on both sides by older rocks dipping toward the core. These relationships are illustrated in Figures 7.4 and 7.13. Determining the sub-surface structure of folds by surface mapping has been an important method for finding oil, as described in the Practicing Geology exercise at the end of the chapter.

Circular Structures

Deformation along plate boundaries by horizontally directed forces usually results in linear faults and folds oriented nearly parallel to the plate boundary. Some types of deformation, however, are more symmetrical, forming nearly circular structures called basins and domes.

A **basin** is a synclinal structure, a bowl-shaped depression of rock layers in which the beds dip toward a central

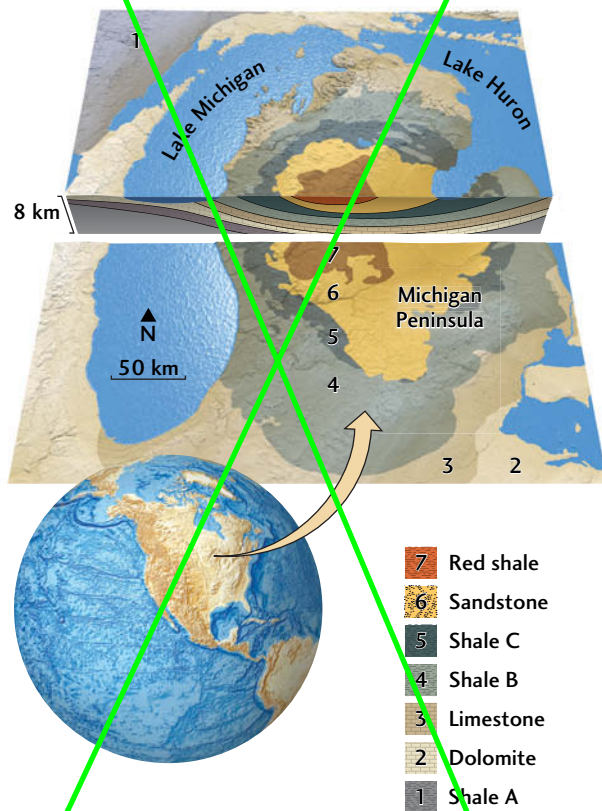


FIGURE 7.15 ■ A geologic map and cross section of the Michigan Basin, which shows sedimentary layers deposited in a thick sequence from the oldest (formation 1) to the youngest (formation 7) during subsidence of the basin. The cross section has been vertically exaggerated by a factor of 5:1.

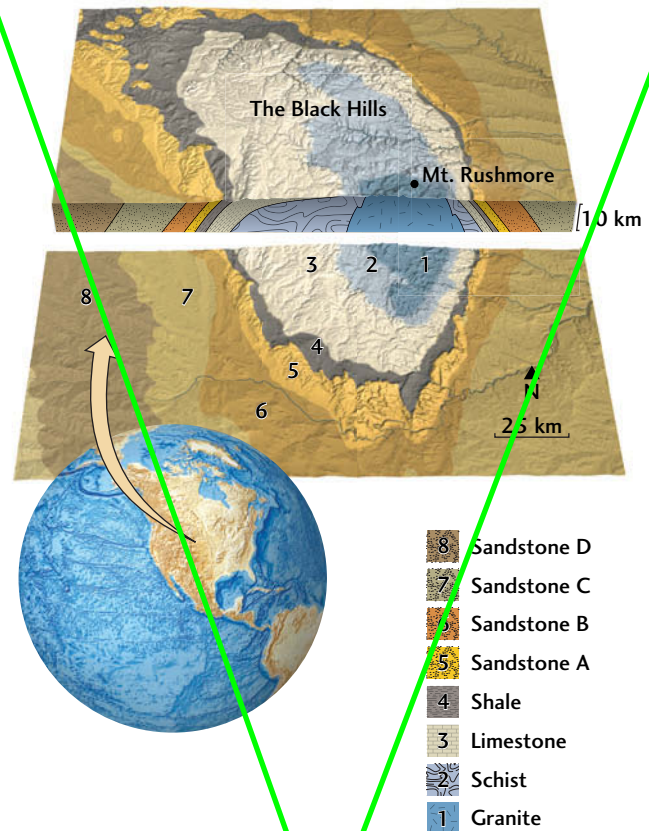


FIGURE 7.16 ■ A geologic map and cross section of the Black Hills dome, which shows sedimentary rocks (formations 3–8) and metamorphic rocks (formation 2) that were uplifted and eroded by the intrusion of a granitic batholith (formation 1). At the Mount Rushmore National Memorial, the faces of four presidents—George Washington, Thomas Jefferson, Theodore Roosevelt, and Abraham Lincoln—are carved into these granitic rocks.

point (Figure 7.15). Sediments are often deposited in basins (see Chapter 5). In some cases, such as the Michigan Basin, shown in Figure 7.15, this deposition can produce sedimentary sequences many kilometers in thickness. A **dome** is an anticlinal structure, a broad circular or oval upward bulge of rock layers. The flanking beds of a dome encircle a central point and dip radially away from it (Figure 7.16). Domes, like other anticlines, are important in petroleum geology because oil is buoyant and tends to migrate upward through permeable rocks (see Practicing Geology exercise). If the rocks at the high point of a dome are impermeable to oil, the oil becomes trapped beneath them.

Domes and basins are typically many kilometers in diameter and some extend for hundreds of kilometers. They are recognized in the field by outcrops that outline their characteristic circular or oval shapes. At these outcrops, the rock layers dip downward toward the center of the basin or upward toward the top of the dome (see Figures 7.15 and 7.16).

Some circular structures are formed by multiple episodes of deformation—for instance, when rocks are compressed in one direction and then again in a direction nearly perpendicular to the original direction. In many other cases, however, these structures result from the upward force of rising material or the downward force of sinking material, rather than the horizontally directed forces of plate tectonics. Not surprisingly, such circular structures tend to be more common in the interiors of plates, far away from active plate boundaries. There are many domes and basins, for example, in the central portion of the United States. Almost all of the Lower Peninsula of Michigan is a large sedimentary basin (see Figure 7.15); the Black Hills of South Dakota are an eroded dome (see Figure 7.16).

Several types of deformation can produce domes and basins. Some domes are formed by rising bodies of buoyant material—magma, hot igneous rock, or salt—that push the overlying sediments upward. As we saw in Chapter 5, some sedimentary basins form when a heated portion of the crust cools and contracts, causing the overlying sediments to subside (thermal subsidence basins). Others result when tectonic forces stretch and thin the crust (rift basins) or compress it downward (flexural basins). The weight of sediments deposited by a river delta can depress the crust into a sedimentary basin, such as the very large basin now forming at the mouth of the Mississippi River in the Gulf of Mexico.

Joints

As we have seen, a fracture that has displaced the rock on either side is called a fault. A second type of fracture is a **joint**—a crack in a rock formation along which there has been no appreciable movement (Figure 7.17a).

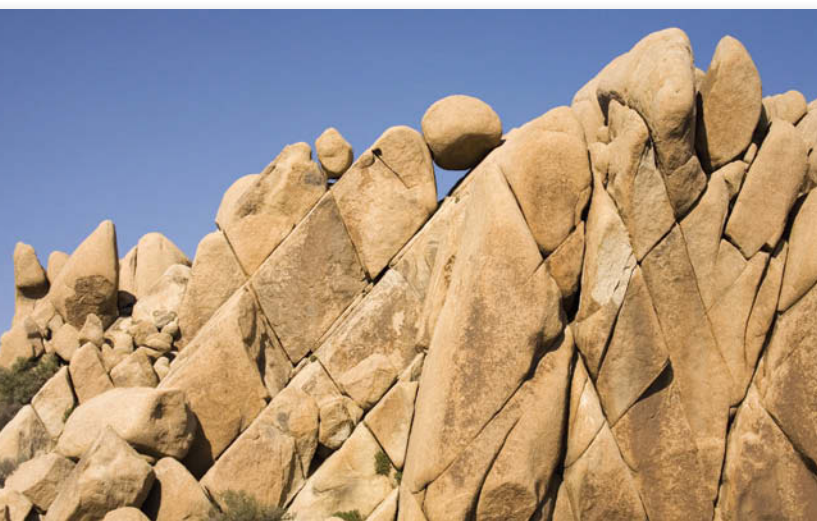
Joints are found in almost every outcrop. Some joints are caused by tectonic forces. Like any other brittle material, brittle rocks subjected to force fracture most easily at flaws or weak spots. These flaws can be tiny cracks, fragments of other materials, or even fossils. Regional tectonic forces—compressive, tensional, or shearing—may leave a set of joints as their imprint long after they have vanished.

The nontectonic expansion and contraction of rock can also form joints. Regular patterns of joints are often found in plutons and lavas that have cooled, contracted, and cracked. Erosion can strip away surface layers, releasing the confining pressure on underlying formations and allowing the rocks to expand and split at flaws.

Joints are usually only the beginning of a series of changes that greatly alter rock formations as they age. For example, joints provide channels through which water and air can reach deep into a formation and speed the weathering and internal weakening of its structure. If two or more sets of joints intersect, weathering may cause the formation to break into large columns or blocks (Figure 7.17b). The circulation of hydrothermal solutions through joints can deposit minerals such as quartz and calcite, forming veins, as we saw in Chapter 3.

Deformation Textures

Joints are examples of small features in rock formations that are best observed up close at an outcrop. Another type of small-scale deformation structure is the texture of a rock mass in areas of localized shearing, such as fault zones.



(a)

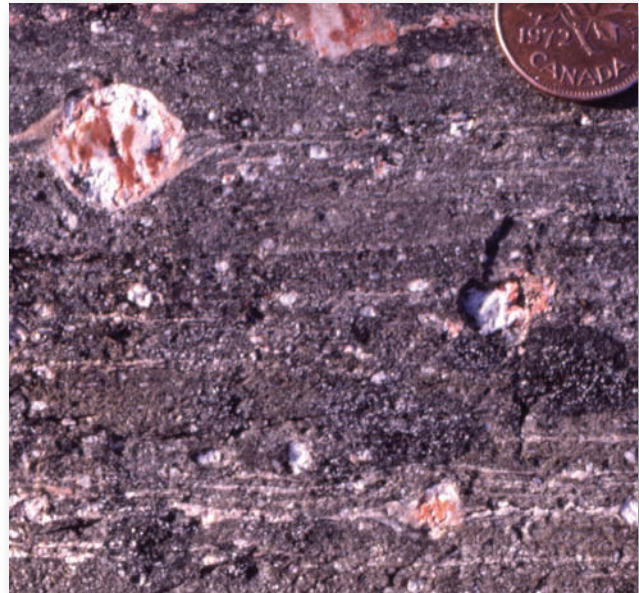


(b)

FIGURE 7.17 ■ Joint patterns. (a) Intersecting joints in a massive granite outcrop, Joshua Tree National Park, California. (b) Columnar joints in basalt, Giants Causeway, Northern Ireland. [(a) Sean Russell/Photolibary; (b) Michael Brooke/Photolibary/Getty Images, Inc.]



(a)



(b)

FIGURE 7.18 ■ (a) Fault breccia developed on a fault in eastern Nevada. The rust-colored breccia shows a cataclastic texture. The gray rocks on either side are limestones. (b) Mylonite developed in the Great Slave Lake shear zone, Northwest Territories, Canada. The rock was originally a granite. As a result of intense shearing forces, the large, originally angular potassium feldspar crystals have been rolled and transformed into smooth balls. [(a) Marli Bryant Miller; (b) Courtesy of John Grotzinger.]

As we have seen, tectonic forces cause the brittle parts of Earth's crust to crack and slip. As the rocks along a fault plane shear past each other, they grind and mechanically fragment solid rock. Where rocks behave as brittle materials (usually in the upper crust), shearing produces rocks with *cataclastic textures*, in which the grains are broken, angular fragments. One such rock type, called *fault breccia*, is shown in **Figure 7.18a**.

Deeper in the crust, where temperatures and pressures are high enough to allow ductile deformation, shearing forces can produce metamorphic rocks called *mylonites* (Figure 7.18b). The movement of rock surfaces against one another recrystallizes minerals and strings them out in bands or streaks. Development of mylonites typically occurs at the greenschist to amphibolite grades of metamorphism (see Chapter 6). The textural effects of deformation are most obvious in mylonites, but they are also prominent in cataclastic rocks.

The San Andreas fault of Southern California makes a good case study of how deformation textures might relate to changes in temperature and pressure with depth. This fault, which marks the boundary between the Pacific Plate and the North American Plate (see Figure 7.7), extends through the crust and probably down into the mantle. At depths up to about 20 km, the fault zone is thought to be very narrow and characterized by cataclastic textures, indicating brittle deformation. Earthquakes are generated in this zone. Below 20 km, however, earthquakes do not occur, and the fault is thought to be characterized by a broad zone of ductile deformation that produces mylonites.

■ Styles of Continental Deformation

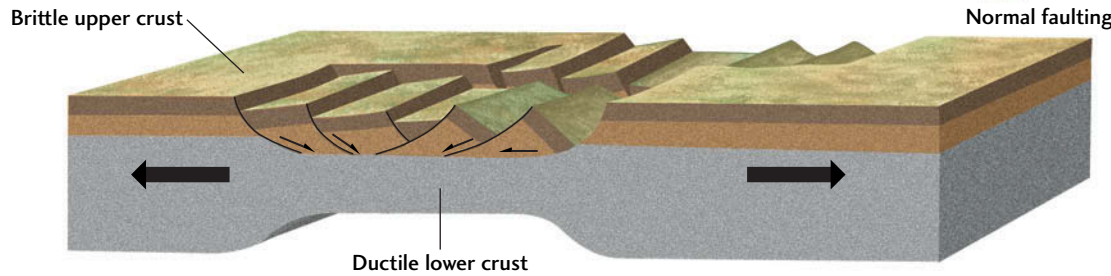
If we look closely enough, we can find all the basic deformation structures—faults, folds, domes, basins, joints—in any zone of continental deformation. But when we view continental deformation at a regional scale, we see distinctive patterns of faulting and folding that relate directly to the tectonic forces causing the deformation. **Figure 7.19** depicts the deformation styles typical of the three main types of tectonic forces.

Tensional Tectonics

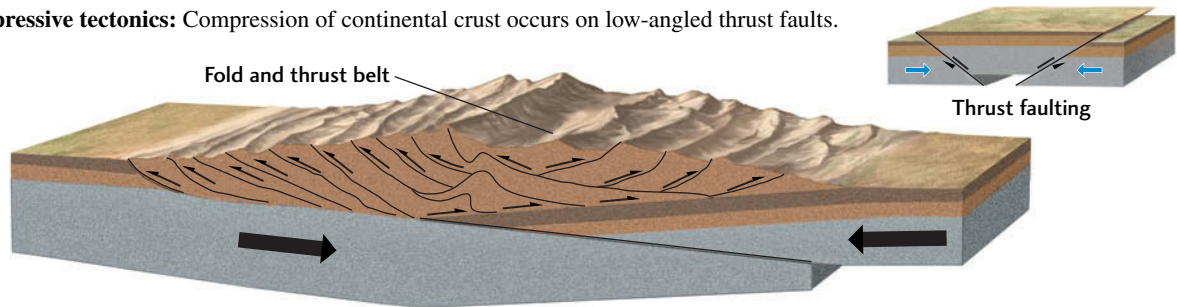
In brittle crust, the tensional forces that produce normal faulting may split a plate apart, resulting in a *rift valley*—a long, narrow trough formed when a block of rock drops downward relative to its two flanking blocks along nearly parallel, steeply dipping normal faults (Figure 7.19a). Well-known examples include the Rhine Valley, the rift valleys of East Africa (**Figure 7.20**), and the Red Sea, as well as the rift valleys of mid-ocean ridges. As we saw in Chapter 5, these structures form basins that fill with sediments eroded from the rift walls as well as with volcanic rocks extruded from tensional cracks in the crust.

Tension on the shallow continental crust usually produces normal faults with high dip angles, typically 60° or

- (a) **Tensional tectonics:** Extension of continental crust produces normal faults with high dip angles in the upper crust that flatten with depth, forming curved fault surfaces.



- (b) **Compressive tectonics:** Compression of continental crust occurs on low-angled thrust faults.



- (c) **Shearing tectonics:** Shearing of continental crust occurs on a nearly vertical strike-slip fault. The case shown here is for a right-lateral fault.

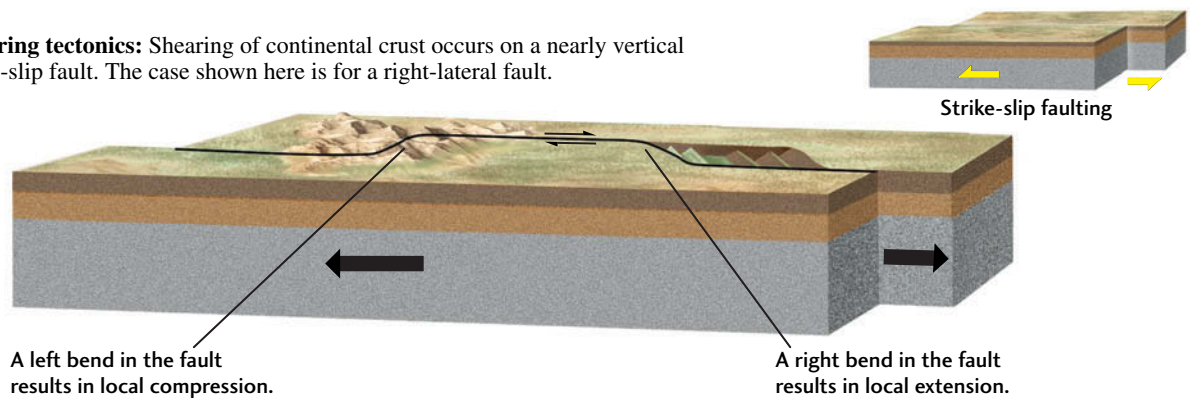


FIGURE 7.19 ■ The orientation of tectonic forces—(a) tensional, (b) compressive, and (c) shearing—determines the style of continental deformation. On a regional scale, the basic types of faulting shown in the inset figures can lead to distinctive, complex patterns of deformation. [After John Suppe, *Principles of Structural Geology*. Upper Saddle River, NJ: Prentice Hall, 1985.]

more. Below a depth of about 20 km, however, crustal rocks are hot enough to behave as ductile materials, and deformation occurs by stretching rather than by fracturing. This change in rock behavior causes the dip of the faults to flatten with increasing depth, which results in normal faults with curved fault surfaces (called *listric* faults), as shown in Figure 7.19a. The crustal blocks moving along these curved faults are tilted backward as the stretching continues.

The Basin and Range province, which is centered on the Great Basin of Nevada and Utah, is a good example of a region defined by many adjacent rift valleys. This region, which is now more than 800 km wide, has been stretched and extended in a northwest-southeast direction by a factor of 2 during the last 15 million years. Here, normal

faulting has created an immense landscape of eroded, rugged fault-block mountains and smooth, sediment-filled valleys, some covered with recent volcanic rocks (see Figure 10.5). This tensional deformation, which appears to be caused by upwelling convection currents beneath the Basin and Range province, continues today.

Compressive Tectonics

In subduction zones, oceanic lithosphere slips beneath an overriding plate along a huge thrust fault, or *megathrust*. The world's largest earthquakes, such as the great Tohoku, Japan, earthquake of March 11, 2011, which generated a disastrous tsunami that killed more than 19,000 people, are

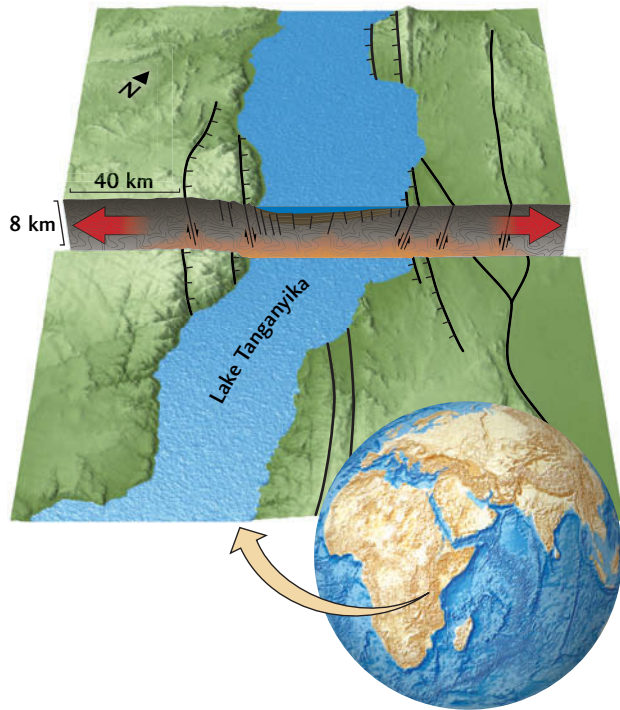
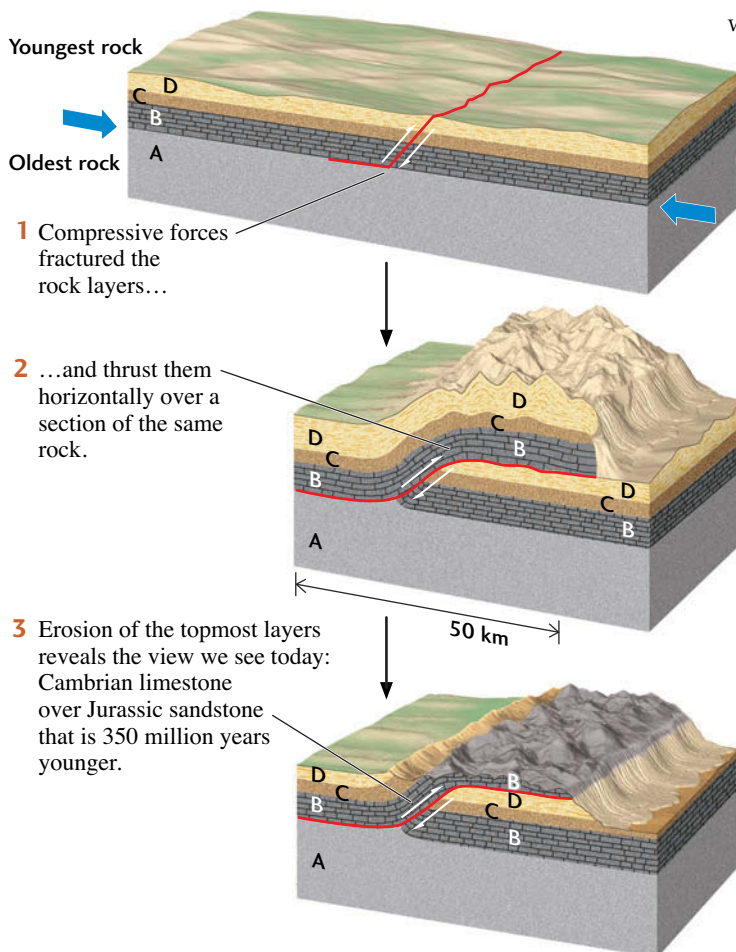


FIGURE 7.20 ■ In East Africa, tensional forces are pulling the Somali subplate away from the African Plate, creating rift valleys bounded by normal faults (see Figure 2.8b). The rift valley shown here is filled by sediments and Lake Tanganyika, on the boundary between Tanzania and the Democratic Republic of Congo. The cross section has been vertically exaggerated by a factor of 2.5:1, which exaggerates the fault dips; the actual dips of the normal faults are about 60°.

caused by sudden slips on megathrusts. Thrust faulting is also the most common type of faulting within continents undergoing tectonic compression. Sheets of crust may glide over one another for tens of kilometers along nearly horizontal thrust faults, forming *overthrust* structures (Figure 7.21).

When two continents collide, the crust can be compressed across a wide zone, resulting in spectacular episodes of mountain building. During such collisions, the brittle basement rocks ride over one another by thrust faulting while the more ductile overlying sedimentary rocks are compressed into a series of great folds, forming a *fold and thrust belt* (Figure 7.19b). Large earthquakes are common in fold and thrust belts; a recent example is the great Wenchuan earthquake that hit Sichuan, China, on May 12, 2008, killing more than 80,000 people.

The ongoing collisions of Africa, Arabia, and India with the southern margin of the Eurasian continent have



Keystone thrust fault, southern Nevada

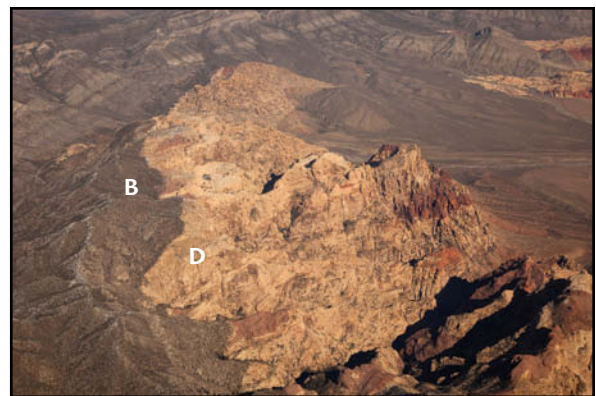


FIGURE 7.21 ■ The Keystone thrust fault of southern Nevada is a large-scale overthrust structure of a kind formed during episodes of continental compression. Compressive forces have detached a sheet of rock layers (D, C, B) and thrust it a great distance horizontally over a section of the same rock layers (D, C, B, A). [Marli Bryant Miller.]

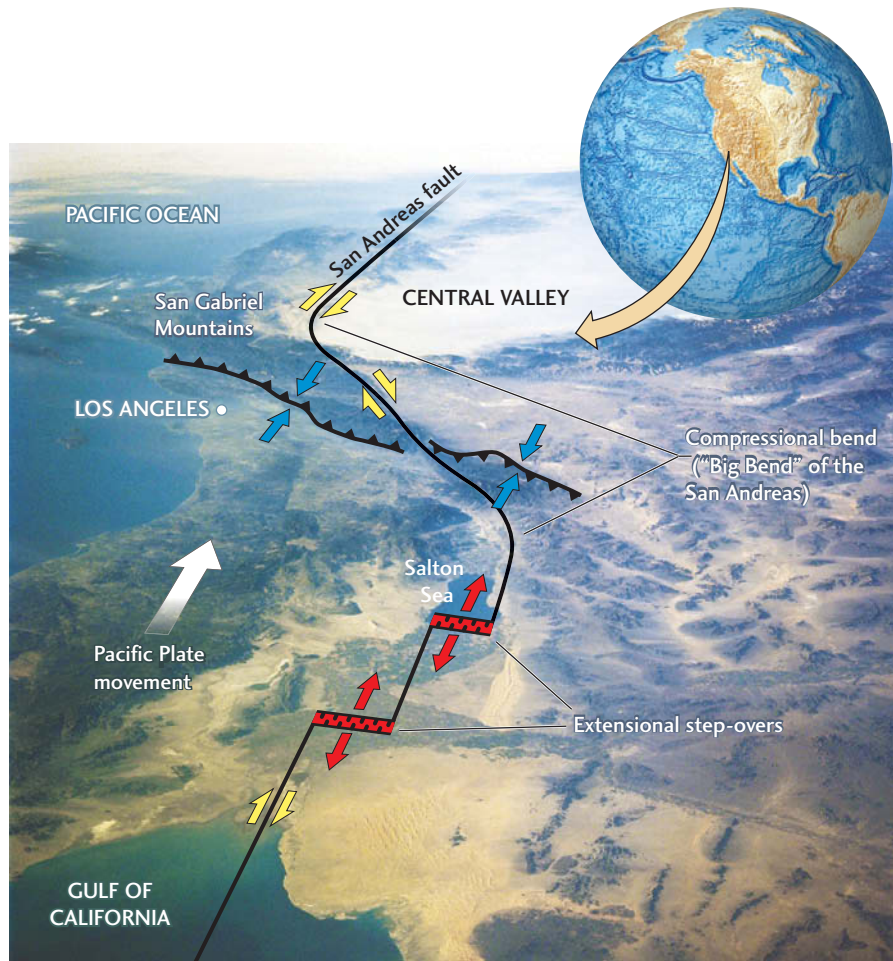


FIGURE 7.22 ■ Space Shuttle photograph showing an oblique view of the San Andreas fault system. The annotations illustrate how the deviations of a transform fault's strike from the direction of plate motion can cause local extension and compression. Between the Gulf of California and the Salton Sea (near bottom of figure), the fault system jogs to the right in two major steps; the right-lateral fault segments (black lines), which are parallel to the Pacific-North America plate motion, are separated by rift valleys (in red) that are volcanically active, subsiding, and filling with sediments. As we look northward, the fault trace first bends to the left, away from the direction of plate motion, and then to the right, realigning with the plate motion in central California (near top of figure). This "Big Bend" in the San Andreas fault causes compression, which is taken up by reverse faulting in the Los Angeles region (middle of figure). [Image Science & Analysis Laboratory, NASA Johnson Space Center.]

created fold and thrust belts from the Alps to the Himalaya, many of which are still active. The great oil reservoirs of the Middle East are trapped in anticlines formed by this deformation. Compression across western North America, caused by that continent's westward movement during the opening of the Atlantic Ocean, created the fold and thrust belt of the Canadian Rockies. The Valley and Ridge province of the Appalachian Mountains is an ancient fold and thrust belt that dates back to the collisions that created the super-continent Pangaea.

Shearing Tectonics

A transform fault is a strike-slip fault that forms a plate boundary. Transform faults such as the San Andreas can offset geologic formations by long distances (see Figure 7.7), but as long as they stay aligned with the direction of relative plate movement, the blocks on either side can slide past each other without much internal deformation. Long transform faults are rarely straight, however, so deformation patterns along these faults can be much more complicated. The faults may have bends and jogs that change the tectonic forces acting across portions of the plate boundary from shearing

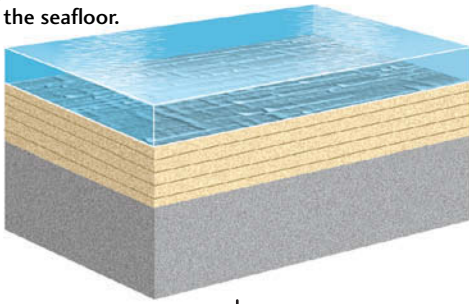
forces to compressive or tensional forces. Those forces, in turn, cause secondary faulting and folding (Figure 7.19c).

A good example of these complications can be found in Southern California, where the right-lateral San Andreas fault bends first to the left and then to the right as one moves along its trace from south to north (Figure 7.22). The segments of the fault on both sides of this "Big Bend" are aligned with the direction of relative plate movement, so the blocks slip past each other there by simple strike-slip faulting. Within the Big Bend, however, the change in the fault orientation causes the plates to push against each other, which produces thrust faulting to the south of the fault. This thrusting has raised the San Gabriel and San Bernardino mountains to elevations exceeding 3000 m and, during the last half century, has produced a series of destructive earthquakes, including the 1994 Northridge quake, which caused more than \$40 billion in damage to Los Angeles (see Chapter 13).

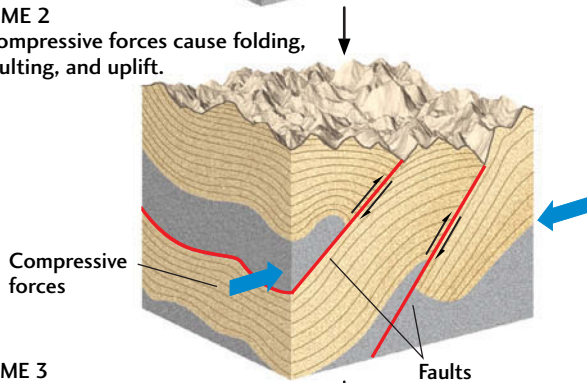
At the southern end of the San Andreas, between the Gulf of California and the Salton Sea, the boundary between the Pacific and North American plates jogs to the right in a series of steps. Within these jogs, the plate boundary is subjected to tensional forces, and normal faulting has

TIME 1

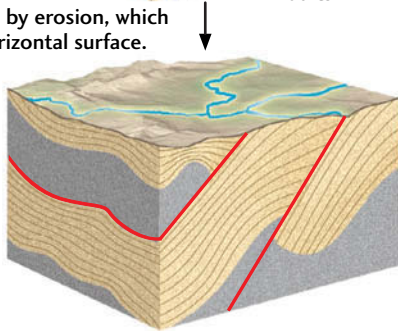
Horizontal layers of sediments are deposited on the seafloor.

**TIME 2**

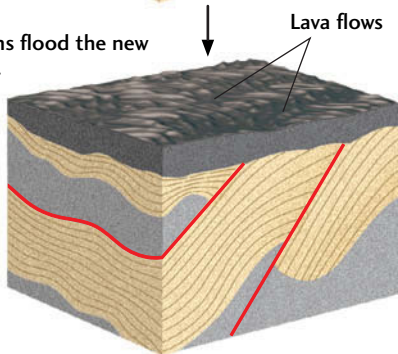
Compressive forces cause folding, faulting, and uplift.

**TIME 3**

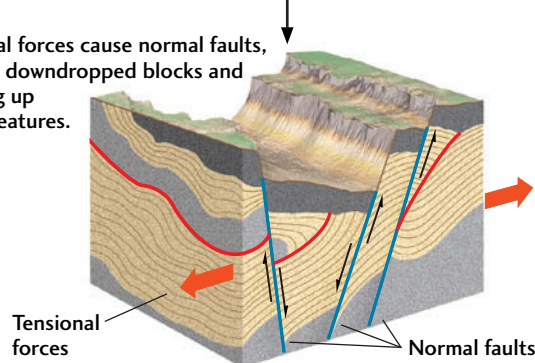
Uplift is followed by erosion, which creates a new horizontal surface.

**TIME 4**

Volcanic eruptions flood the new surface with lava.

**TIME 5**

Tensional forces cause normal faults, creating downdropped blocks and breaking up earlier features.



formed small rift valleys that are volcanically active, rapidly subsiding, and filling with sediments. This extension occurs within 200 km of the Big Bend compression, demonstrating how variable the tectonics along continental transform faults can be!

■ Unraveling Geologic History

The geologic history of a region is a succession of episodes of deformation and other geologic processes. Let's see how some of the concepts and methods introduced in this chapter can be used to reconstruct that history.

The cross sections in [Figure 7.23](#) represent a few tens of kilometers of a geologic region that underwent a succession of tectonic events. First, horizontal layers of sediment were deposited on the seafloor. Those layers were tilted and folded, and eventually uplifted above sea level, by horizontal compressive forces. There, erosion gave them a new horizontal surface. That surface was covered by lava when forces deep in Earth's interior caused a volcanic eruption. In the most recent stage, tensional forces resulted in normal faulting, which broke the crust into blocks.

Geologists see only the last stage, but visualize the entire sequence. They begin by identifying and determining the ages of the rock layers and recording the orientation of layers, folds, and faults on geologic maps. Then they use those maps to construct cross sections of the subsurface features. Once the geologists have identified sedimentary beds, they can start with the knowledge that the beds must originally have been horizontal and undeformed at the bottom of an ancient ocean. The succeeding events can then be reconstructed.

Present-day surface topography in young mountain ranges—such as the Alps, the Rocky Mountains, the Pacific Coast Ranges, and the Himalaya—can be traced in large part to deformation that has occurred over the past few tens of millions of years. These young systems still contain much of the information that geologists need to piece together the history of that deformation. Deformation that occurred hundreds of millions of years ago is no longer evident in the form of rugged mountains, however. Erosion has left behind only the remnants of folds and faults, expressed as low ridges and shallow valleys. As we will see in Chapter 10, even older episodes of mountain building are evident only in the twisted, highly metamorphosed formations that constitute the basement rocks of the interiors of continents.

FIGURE 7.23 ■ Stages in the development of a geologic region. A geologist sees only the last stage and attempts to reconstruct all the earlier stages from the observable structural features.